Mathematics - Graduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the minimum number of people needed in a room to guarantee that there are 4 mutual friends or 4 mutual strangers?

    (a) 18
    (b) 14
    (c) 10
    (d) 25
    (e) 30

2. Suppose f is analytic on the closed unit disk, f(0) = 0, and |f(z)| $\$\leq |\mathrm{e}\hat{}$z| whenever |z| = 1. How big

    (a) 2.3456
    (b) 2.2360
    (c) 3.0000
    (d) 2.9876
    (e) 1.9221

3. Evaluate $\$\int_c z^2/(z-5)dz$, where c is the circle that $|z|=2$.

    (a) $2\pi i$
    (b) 0
    (c) $-2\pi i$
    (d) 1
    (e) $4\pi i$

4. A coach rounded the number of runners at a track meet to the nearest 10. The rounded number of runners is 400. Which number could be the actual number of runners at the track meet?

    (a) 411
    (b) 382
    (c) 420
    (d) 447
    (e) 375

5. Statement 1 | 4x - 2 is irreducible over Z. Statement 2 | 4x - 2 is irreducible over Q.

    (a) True, Not enough information

(b) Not enough information, False

(c) False, False

(d) Not enough information, Not enough information

(e) Not enough information, True

True / False (10 marks)

Answer True or False

6. True or False: For a 2 * 2 matrix A to satisfy AM = MD for some nonzero matrix M, the eigenvalue

7. True or False: The greatest common factor of 252 and 96 is 12.

8. True or False: The limit of the sequence a_n as n approaches infinity is 0.5, given the conditions $0 <$ a_n $<$ 1 and (1-a_n)a_{n+1} $>$ 1/4.

9. True or False: The correct answer to 'x=0.3168. what is the value of $x*$\prod_{gjc} f\infty(1-\frac{r}{afk\pi^2} bwxpzpnynkegahbpey$

9. True or False: The set of nonzero integers under multiplication does not satisfy the group axioms

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the probability of rolling a 6 on a fair 6-sided die across 5 trials. Analyze the conditions under which a 6 appears at most twice, and derive the corresponding probability.

12. Calculate the quantity of a drug in a patients body after taking three doses, given that 5% remains in the body before each daily 150 mg dose. Explain your reasoning and calculations.

End of Paper